cheesecracker

The API server of the Cheese App, built from an empty folder.

- Tests: unit, integration and system
- Checks: coverage, ruff and pre-commit
- Docker: a run image and a dev image

The tool uv installs Python and the packages of the project. The tool just runs each task by a short name. Everything except the Docker images runs on this machine with these two tools.

Unit tests

A unit test calls one function alone, with no server and no network.

Tool	Does
`uv add --group test pytest`	adds pytest to the group `test`
`just test`	runs `uv run pytest`

The API server

FastAPI turns Python functions into answers to HTTP requests. uvicorn runs the app.

Request	Answer
`/`	a welcome
`/health`	`{"status": "healthy"}`

`just serve` starts it in the background at `http://localhost:9000`, and it restarts on each change. `just stop` stops it.

The testing pyramid

Kind	Calls	Needs
Unit	one function	nothing
Integration	the app, through `TestClient`	no server
System	the running server, over HTTP	a server at port 9000

Unit tests are the base of the pyramid, many and fast. System tests are its top, few and slow.

The formatter and the linter

ruff formats the code and checks it against rules. It does the work of two older tools, black, a formatter, and flake8, a linter.

Recipe	Runs	Fixes
`just fmt`	`ruff format`	spaces, blank lines, line breaks
`just lint`	`ruff check`	nothing alone. `just lint --fix` fixes some rules
`just check`	format, lint, tests	nothing: it only checks

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Checks before every commit

pre-commit reads `.pre-commit-config.yaml`, and runs its checks before each `git commit`.

- pre-commit-hooks: whitespace, YAML, JSON, large files, private keys
- ruff-pre-commit: `ruff-check` and `ruff-format`

`just hook` installs the hook, once in each clone.